MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, HIGH-VOLTAGE RECTIFIER TYPES 1N5181, 1N5182, 1N5183, 1N5184 and TX 1N5181, TX 1N5182, TX 1N5183, TX 1N5184

1. SCOPE

1.1 Scope.- This specification covers the detail requirements for silicon, high-voltage, semiconductor diode for use as rectifier devices in compatible equipment circuits. (See 6.2 herein.) The prefix "TX" identifies devices meeting the special process-conditioning, testing, and screening requirements in 4.4 herein, and the stringent quality-control (LTPD) requirements in Tables I, II, and III herein.

1.2 Outline and dimensions.- See Figure 1 herein.

1.3 Operating characteristics and ratings.- (At T_A = +25°C, unless otherwise specified. See 3.4 herein.):

	V_R		I_o 1/				$i_{f(surge)}$		T_{stg}		$T_{(oper)}$	
			at: T_A=+25°C		at: T_A=+150°C		at: 1/120 sec.					
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
	Vdc	Vdc	mAdc	mAdc	mAdc	mAdc	A	A	°C	°C	°C	°C
1N5181	---	4,000	---	125	---	20	---	1	-65	+175	-65	+175
1N5182	---	5,000	---	125	---	20	---	1	-65	+175	-65	+175
1N5183	---	7,500	---	125	---	20	---	1	-65	+175	-65	+175
1N5184	---	10,000	---	125	---	20	---	1	-65	+175	-65	+175

1/ For temperatures (T_A) up to +175°C, derate in accordance with Figure 2 herein.

1.3.1 Operation at high altitudes.- See 3.3.3 and Figure 3 herein.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-19500	Semiconductor Devices, General Specification For
MIL-STD-15	Electrical And Electronic Symbols
MIL-STD-202	Test Methods For Electronic and Electrical Component Parts
MIL-STD-750	Test Methods For Semiconductor Devices

(Copies of specifications, standards, drawing, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Requirements.- Requirements for the semiconductor diodes shall be in accordance with Specification MIL-S-19500 and as otherwise specified herein.

3.2 Abbreviations and Symbols.- The abbreviations and symbols used herein are defined in Specification MIL-S-19500.

3.3 Design and construction.- The semiconductor diodes shall be of the design, construction, and physical dimensions specified in Figure 1 herein.

3.3.1 Terminal arrangement.- The terminal arrangement on the diodes shall be as indicated in Figure 1 herein.

3.3.2 Terminal-lead length.- Terminal-lead length(s) other than that specified in Figure 1 may be furnished under contract or order (see 6.3 herein) where the devices covered herein are required directly for particular equipment-circuit installation. Where such other lead lengths are required and provided, it shall not be construed as affecting adversely the Qualified-product status of the device, or applicable JAN marking.

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3.3.3 Altitude-operation factors.- The semiconductor diodes shall be designed for proper operation (see 3.4 herein), without external shielding or other methods of electrostatic or electromagnetic protection, up to the maximum altitude-and-voltage combinations indicated in Fig. 3 herein.

3.3.4 Polarity indication.- Polarity shall be indicated on the body of the device, at the cathode-terminal end, by means of a contrasting color band or conventional diode symbol. The graphic symbol for polarity shall be as established in Standard MIL-STD-15. Color coding shall not be employed on the semiconductor device.

3.3.5 Operating position.- The diodes shall be capable of proper operation in any position.

3.3.6 Use of material other than glass or metal on device body. Plastic or other molding compound shall be used in the external-body construction of the device and the special test requirements of 4.3.3 shall apply.

3.4 Performance characteristics.- The semiconductor diode performance characteristics shall be as specified in Tables I, II, and III herein (see 3.4.1 below). Except where specifically differentiated for respective, individual diode types (see Tables I, II, and III herein), the performance requirements including characteristics, ratings, and test conditions, apply equally to all diode types covered herein. (See 4.3.3 and 4.4 herein.)

3.4.1 "TX" special requirements.- Specified "TX" device-types shall, in addition to all other performance requirements herein, be process-conditioned, tested, and screened in accordance with 4.4 herein. (At option of the manufacturer, the non-TX device-types may be subjected to the same process-conditioning and testing. See 3.5.1 herein.)

3.5 Marking.- Unless otherwise specified herein, marking shall be in accordance with Specification MIL-S-19500. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the "classification" type-designation only. The "manufacturer's identification" and "country of origin" may, at option of the manufacturer, be omitted from being marked directly on the semiconductor device covered herein.

3.5.1 "TX" identification marking.- Marking of devices conforming to "TX" and all other applicable requirements herein (see 3.4.1), shall include the prefix "TX" immediately preceding the type designation.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

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4.2 Qualification and acceptance inspection.- Qualification and Quality Conformance inspection shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein (see 4.2.2 herein). Groups A, B, and C inspection shall consist of the examinations and tests specified in Table I, II and III, respectively, herein. Quality Conformance inspection shall include inspection of Preparation for Delivery (see 5.1 herein).

4.2.1 Specified LTPD for subgroups.- The LTPD specified for a subgroup in Tables I, II, and III herein shall apply for all of the tests, combined, in the subgroup.

4.2.2 Discriminate Qualification testing.- Only the non-TX device types shall be considered subject to Qualification tests. It shall be understood that procurement of the TX device types will be made from sources for the approved non-TX devices as listed in Qualified Products List (QPL)- 19500. Hereto, upon request and validation by the qualifying activity, qualification may be extended to (and appropriate QPL-listing effected) TX devices manufactured by approved sources for the non-TX devices.

4.2.2.1 Group A and Life Test criteria and requirements for Qualification inspection.-

a. For Qualification inspection, only one (1) failure will be permitted for all Group A tests combined.
Upon occurrence of one such failure, the Chief, Components And Materials Standardization Division, Procurement And Production Directorate, Fort Monmouth, New Jersey 07703, shall be notified immediately.
Further Qualification inspection shall not be continued until so authorized from the above mentioned Government agency.

b. For Qualification inspection, continuous required (total-hours duration--through Group C requirement thereto) Life Test(s) shall be effective. (See 4.2.4 herein.)

4.2.3 Acceptance inspection of lots containing multiple diode types.- For lots containing more than one diode type covered herein, subjection of adequate samples on a sublot basis to tests in accordance with a, b, and c, below, will be acceptable. Normally applicable acceptance criteria shall apply to acceptability determination for the entire submitted lot.

a. Each diode type (sublot) shall be subjected to all Group A tests and to all Group C tests (when the Group C tests are incumbent to be performed on the diode type concerned); plus:

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b. A sublot, selected by the Government inspector, and containing as equal as possible quantities of the diode types involved for lot acceptance, shall be subjected to all Group B, Subgroups 1 through 4 tests; plus:

c. A sublot consisting of the highest-voltage type involved for lot acceptance shall be subjected to Group B, Subgroup 5 and 6 test(s).

4.2.4 Group B-Group C life test samples.- Samples that have been subjected to Group B, 340-hour life test may be continued on test for 1000 hours in order to satisfy Group C life test requirements. These samples shall be predesignated, and shall remain subjected to the Group C 1000-hour evaluation after they have passed the Group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1000 hours shall be computed for 1000-hour acceptance criteria. (See 4.2.5 below.)

4.2.5 Group C testing.- Group C tests shall be performed on a lot every 6 months. (See Table III herein.) The contractor shall, throughout the course of a contract or order, permit the Government representative to scrutinize all test data and findings covering manufacturer's test program on Group C characteristics and parameters for the product concerned. Upon determination by the Government inspector (in advance of Group C, 6-month, test results) that Group C parameters are not being adequately met, the Government inspector may require lot-by-lot inspection, normally for a minimum of 3 consecutive lots, to be performed for required Group C tests.

4.2.6 Disposition of sample units.- Sample units that have been subjected to Group C, Subgroup 1 and 2 tests shall not be delivered on the contract or order. Sample units that have been subjected to and have passed Group B, Subgroups 2 through 6 tests and Group C, Subgroup 3, 4, and 5 tests, (these tests to be considered non-destructive), may be delivered on the contract or order provided that, after Group B and C inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

4.2.7 Copies of lot inspection data.- When specified in the contract or order, one copy of the Quality Conformance inspection data, including manufacturer's test data and findings, covering the device inspection lot being shipped shall accompany the device(s) concerned.

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4.3 Particular examination and test requirements.-

4.3.1 Interval for End-Point test measurements.- All applicable End-Point Test measurements shall be performed within 96 hours after sample units have been subjected to required physical-mechanical or environmental test(s).

4.3.2 Mechanical damage resulting from tests.- Except for intentionally deforming, mutilating, or dismembering mechanical-stress tests to which samples are subjected, there shall be no evidence of mechanical damage to any sample unit as a result of any of the Group A, B, or C tests.

4.3.3 Special testing requirements for non-glass or non-metal (body contruction) devices only (see 3.3.6 herein).- The following test(s) and special test conditions shall apply thereto:

a. Moisture Resistance (Group B test)-- all units under test shall be energized, throughout the test, at V_F = 100 Vdc.

b. Storage Life (Group B and C High-Temp. Life Non-Oper. test)-- all units under test shall be energized, throughout the test at V_F = 200 Vdc.

c. Flammability (Group C test)-- point of impingement of flame shall be as indicated in Figure 1 herein; time of burning of visible flame on unit under test shall not exceed 5 seconds (after removal of applied flame); no violent burning or explosive-type fire shall be manifested on the unit subjected to test. During an interval not exceeding 30 hours after Flammability test, the test units shall be subjected to End-Point Tests as follows:

(1) Rated V_R (V_R reading shall be taken within 5 seconds of energization of device), and I_R test as specified for Subgroup 2 in Group A inspection herein, except:

(2) I_R test limit may be exceeded by not more 50%.

Sample units subjected to Flammability test shall not be shipped on contract or order.

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4.3.4 Surge Current test.- The specified test current, I, shall be applied in the breakdown region, at the temperature (T_C) specified, and there shall be superimposed on that current the specified surge current, i_f(surge), for a total of ten surges at one-minute intervals. Each individual surge shall be a 1/2 square-wave pulse, or equivalent sine-wave pulse, of 1/120-second duration. (Method 4066 in Standard MIL-STD-750 may be referred to for additional guidance for the test.)

4.4 Process conditioning, testing, and screening for "TX" devices.- All "TX" devices (100% of each lot) shall be subjected to the process conditioning, testing, and screening specified in 4.4.2 through 4.4.6.4 herein, in accordance with the procedural sequence established in Figure 4 herein, prior to submission of the lot to Group A, B, C tests. Those conditioning and screening tests normally performed by a manufacturer as standard-production tests, need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than those specified herein.

4.4.1 Quality Assurance records and determinations.- The manufacturer shall maintain and make available for perusal and review at any time, during a period of 3 years minimum, to the authorized Government representative or other contracting customer concerned, complete lot-by-lot records for these devices. All devices that fail to meet the process-conditioning, testing, and screening requirements specified herein shall be removed from the respective lot, and the test-failure identity and quantity removed shall be recorded in the permanent lot history. (See 6.3, 6.4, and 6.5 herein.)

4.4.2 High-temperature storage.- The diodes shall be stored for at least 24 hours at a minimum temperature (Tstg) of +200°C.

4.4.3 Thermal shock (temperature cycling).- The diodes shall be subjected to (thermal shock) temperature cycling in accordance with Standard MIL-STD-750, Method 1051, Test Condition C, except that $T_{(high)}$ shall be +175°C and the cycling duration shall be 10 continuous cycles and the exposure time at the temperature extremes shall be 15 minutes, minimum.

4.4.4 Acceleration.-The diodes shall be subjected to the constant acceleration (centrifugal) test per Method 2006 in Standard MIL-STD-750, except that the following shall apply:

a. Centrifugal force = 1,000 G.

b. Orientation = Y1, only.

c. Hold time of 1 minute shall not apply hereto.

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4.4.5 Seal (fine-leak) test.- The diodes shall be subjected to the Seal test per Standard MIL-STD-202, Method 112, Test Condition C, procedure III a or III b (see 4.4.5.1 or 4.4.5.2 below), except that testing for gross leaks shall be in accordance with 4.4.5.3 below.

4.4.5.1 Conditions for procedure IIIa.- The sealed chamber in which the diodes are placed shall be pressurized with helium gas to 50 psig, minimum, for a minimum of 4 hours. At end of that period, the diodes shall be removed from the chamber and, within 30 minutes, be subjected to a helium leak-detection test. Those diodes showing a leakage rate of greater than 1×10^{-7} cc/sec of helium, when measured at a differential pressure of 1 atmosphere, shall be rejected and removed from the lot.

4.4.5.2 Conditions for procedure IIIb.- The sealed chamber in which the diodes are placed shall be pressurized to 30 psig, minimum, for a minimum of 6 minutes, with Krypton 85 tracer in a nitrogen solution. Within 4 hours after evacuation of the chamber, the diodes shall be tested with a radioactivity counter capable of detecting leakage at the rate of 1×10^{-8} cc/sec. Those diodes showing that leakage rate or greater shall be rejected and removed from the lot.

4.4.5.3 Gross Leak test.- The diodes shall be immersed for a minimum period of 15 seconds in non-corrosive ethylene glycol, maintained at approximately 100°C, and shall be observed for emanating bubbles. Those diodes from which a bubble trace is observed shall be rejected and removed from the lot.

4.4.6 Burn-in conditioning, test, and evaluation (screening).- To enable proper evaluation of the diodes conditioned and tested in accordance with the following requirements, each diode and lot shall be handled or identified in such manner that results for each individual diode and lot can be ascertained and recorded. (See 4.4.6.4 herein.)

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4.4.6.1 Pre burn-in measurements.- The diodes shall be tested for the characteristics listed in Table IV below, and the values found shall be recorded. Those diodes that fail to meet the specified limits shall be rejected and removed from the lot.

Table IV. Pre burn-in measurements.

Test Method per MIL-STD-750	Examination or test 1/	Conditions	Symbol	Limits Min	Limits Max	Unit
4011	Forward voltage	I_F=0.125 Adc	V_F	---	10	Vdc
4016	Reverse current:	D.C. Method				
	1N5181	V_R=4,000 Vdc	I_R	---	1.0	uAdc
	1N5182	V_R=5,000 Vdc	I_R	---	1.0	uAdc
	1N5183	V_R=7,500 Vdc	I_R	---	1.0	uAdc
	1N5184	V_R=10,000 Vdc	I_R	---	1.0	uAdc

1/ See 3.4 herein.

4.4.6.2 Burn-in conditioning.- The diodes shall be energized for 96 hours, minimum, at I_F = 0.125 Adc, minimum.

4.4.6.3 Post burn-in tests and measurements.- After burn-in conditioning, the diodes shall be permitted to return to room-ambient temperature thermal-equilibrium and retested, within a maximum total elapsed time of 96 hours, for the characteristics listed in Table IV above. The values now found for each individual diode shall not exceed the following (respectively applicable to the pre burn-in measurements per Table IV above):

ΔV_F = +20% change, max.

ΔI_R = +400% change, max.

4.4.6.4 Burn-in conditioning evaluation (screening). Those diodes whose measured characteristics exceed the delta (Δ) limits specified in 4.4.6.3, above, shall be rejected and removed from the lot. Where 10% or more of the diodes are found to exceed those limits, the entire lot shall be considered non-acceptable as "TX" devices.

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Table I. Group A inspection.

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		5	3				
2071	Visual and mechanical examination	---			---	---	---	---
	Subgroup 2		5	3				
4011	Forward voltage	I_F=0.125 Adc			V_F	---	10	Vdc
4011	Forward voltage (at high temp.)	I_F=0.060 Adc T_A = +100°C 2/			V_F	---	10	Vdc
4016	Reverse current:	D.C. Method						
	1N5181	V_R=4,000 Vdc			I_R	---	1.0	uAdc
	1N5182	V_R=5,000 Vdc			I_R	---	1.0	uAdc
	1N5183	V_R=7,500 Vdc			I_R	---	1.0	uAdc
	1N5184	V_R=10,000 Vdc			I_R	---	1.0	uAdc
4016	Reverse current (at high temp.):	D.C. Method T_A=+100°C 2/						
	1N5181	V_R=4,000 Vdc			I_R	---	20	uAdc
	1N5182	V_R=5,000 Vdc			I_R	---	20	uAdc
	1N5183	V_R=7,500 Vdc			I_R	---	20	uAdc
	1N5184	V_R=10,000 Vdc			I_R	---	20	uAdc

1/ See 3.4 herein.

2/ Test measurement shall be made after thermal equilibrium has been reached at the temperature specified.

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Table II. Group B inspection.

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		10	5				
2066	Physical dimensions	---			---	---	---	---
	Subgroup 2		10	5				
1051	Temperature cycling	Test Cond. C except $T_{(high)}$=+175°C			---	---	---	---
1056	Thermal shock (glass strain)	Test Cond. A			---	---	---	---
1021	Moisture resistance	No initial conditioning 2/			---	---	---	---
2036	Terminal strength, (lead fatigue)	Test Cond. E			---	---	---	---
	End-point tests: Group A, Subgroup 2 tests and limits apply							
	Subgroup 3		10	5				
2016	Shock	Non-operating 1500 G 5 blows of 0.5 msec ea. in orientations X1, Y1 Z1 (total = 15 blows)			---	---	---	---
2046	Vibration fatigue	Non-operating			---	---	---	---
2056	Vibration, variable frequency	---			---	---	---	---

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Table II. Group B inspection-(Cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 3 -(Cont'd)							
2006	Constant acceleration	1,000G Orientations X1, Y1, Z1			---	---	---	---
	End-Point Tests: Group A, Subgroup 2 tests and limits apply							
	Subgroup 4		10	5				
3/	Surge current	I=125 mAdc i_f(surge)=1.0 Adc T_C=+100°C			---	---	---	---
	End-point tests: Group A, Subgroup 2 tests and limits apply							
	Subgroup 5		10	λ=7				
1031	High-temperature life (non-operating)	T_{stg} =+150°C 4/; 5/ t =340 hours 6/			---	---	---	---
	End-Point tests: Group A, Subgroup 2 tests and limits apply							

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Table II. Group B inspection-(Cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 6		10	λ=7				
1026	Steady state operation life:	T_C=+100°C I = 60 mAdc 4/ t=340 hrs. 6/				---	---	---
	1N5181	V_R=4,000 Vdc						
	1N5182	V_R=5,000 Vdc						
	1N5183	V_R=7,500 Vdc						
	1N5184	V_R=10,000 Vdc						
	End-point tests: Group A, Subgroup 2 tests and limits apply							

1/ See 3.4 and 4.3.1 herein.

2/ See 4.3.3 a herein.

3/ See 4.3.4 herein.

4/ See 4.2.4 herein.

5/ See 4.3.3 b herein.

6/ For non-"TX" devices only. "TX" devices shall be subjected to 1000-hour test duration.

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Table III. Group C inspection. 1/

Test Method per MIL-STD-750	Examination or test 2/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		10	5				
1001	Barometric pressure, reduced (altitude operation):	Pressure=8.8 in. Hg Normal mounting t=1 min., minimum						
	1N5181	V_R=4,000 Vdc						
	1N5182	V_R=5,000 Vdc						
	1N5183	V_R=7,500 Vdc						
	1N5184	V_R=10,000 Vdc						
1046	Salt-spray (corrosion)	Test Cond. A			---	---	---	---
	End-point tests: Group A, Subgroup 2 tests and limits apply							
	Subgroup 2 3/		15	10				
4/	Flammability	5/			---	---	---	---
	Subgroup 3		15	10				
6/	Seal (leak rate)	---			---	---	10^{-7}	atm cc/sec
	Subgroup 4		λ=10		---			
1031	High-temperature life (non-operating)	T_{stg}=+150°C 7/; 8/			---	---	---	---
	End-point tests: Group A, Subgroup 2 tests and limits apply							

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Table III. Group C inspection. 1/ -(Cont'd).

Test Method per MIL-STD-750	Examination or test 2/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 5		λ=10	---				
1026	Steady-state operation life:	T_C=+100°C I = 60 mAdc 7/			---	---	--	--
	1N5181	V_R=4,000 Vdc						
	1N5182	V_R=5,000 Vdc						
	1N5183	V_R=7,500 Vdc						
	1N5184	V_R=10,000 Vdc						
	End-point tests: Group A, Subgroup 2 tests and limits apply							

1/ See 4.2.5 herein.

2/ See 3.4 and 4.3.1 herein.

3/ See 4.3.3 herein.

4/ Per Method 111 in Standard MIL-STD-202

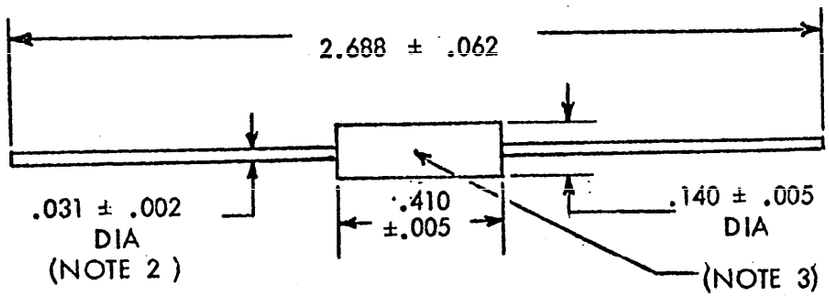
5/ See 4.3.3 c herein.

6/ See 4.4.5 herein.

7/ See 4.2.4, and subnotation 6/ in Table II, herein.

8/ See 4.3.3 b herein.

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NOTES:

1. All dimensions in inches.
2. Solid silver leads.
3. Locale for impingement of flame during Flammability test.

Figure 1. Outline and dimensions.

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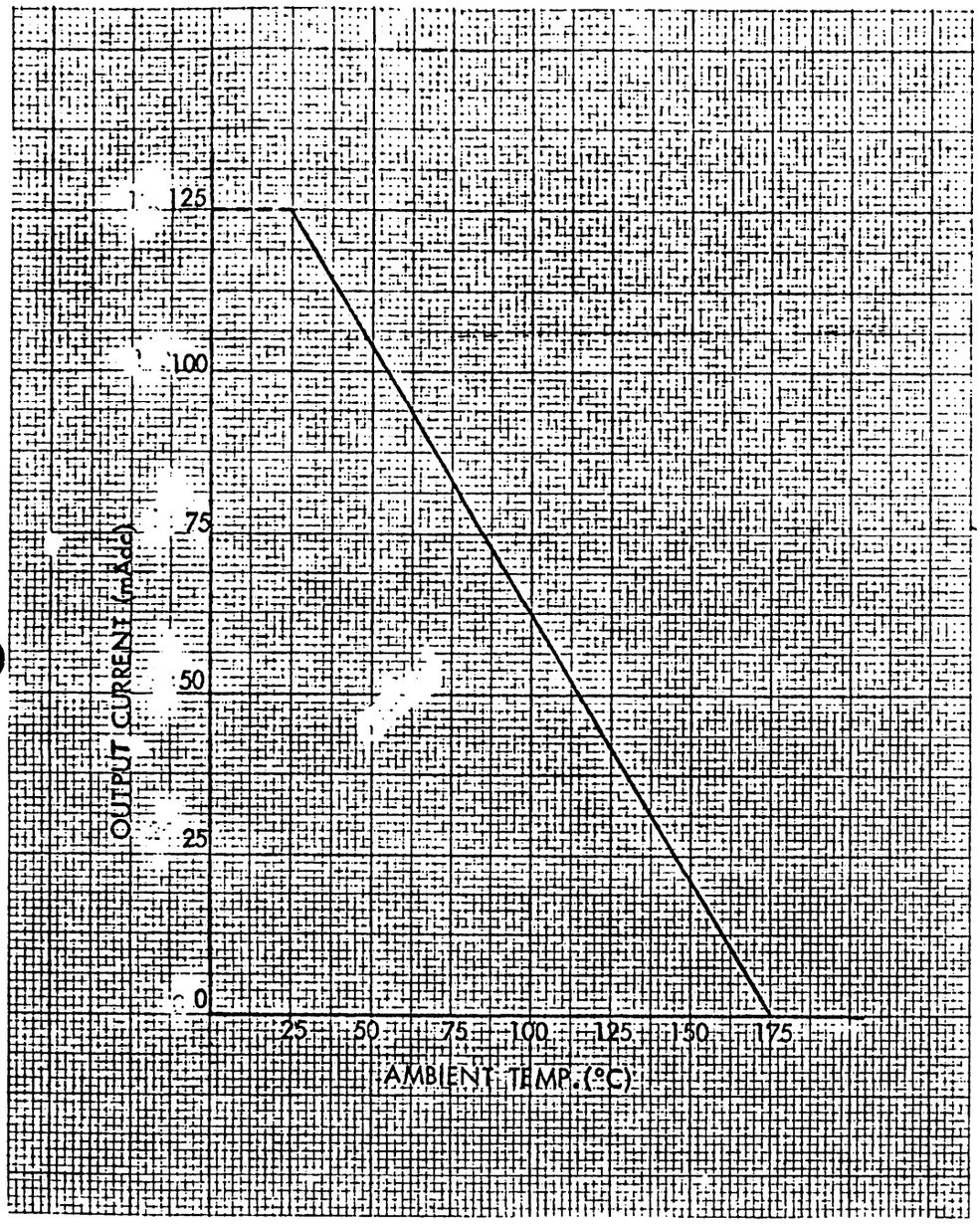


Figure 2. Current-vs-Temperature Derating nomograph.

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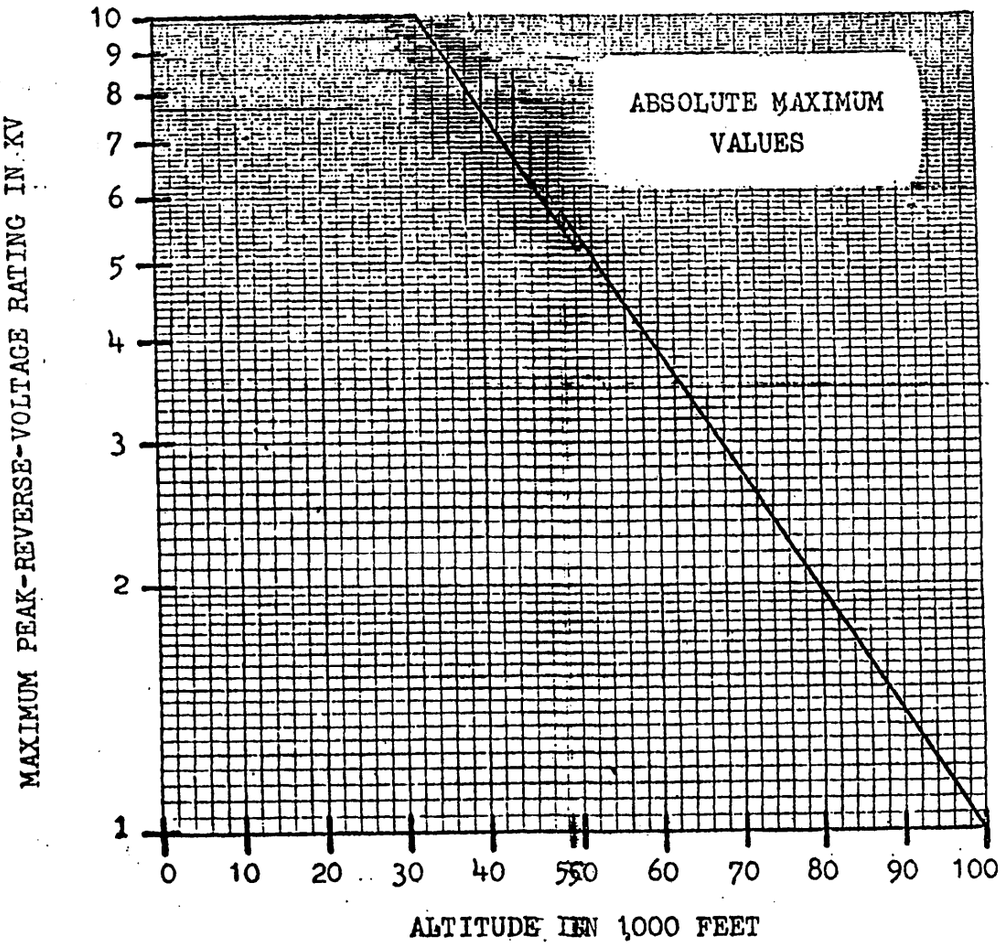


Figure 3. Voltage-vs-Altitude Operation derating nomograph.

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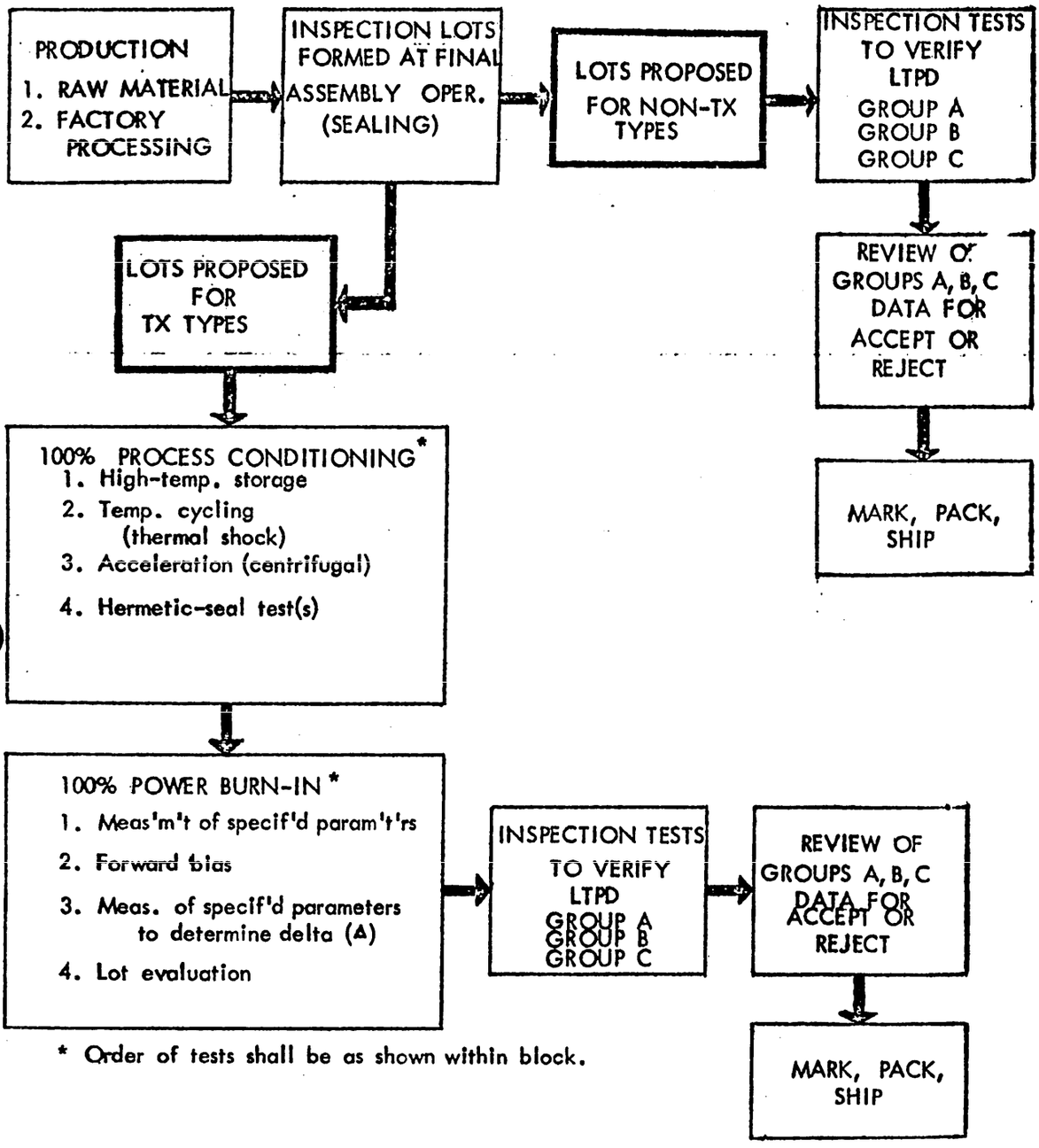


Figure 4. Flow chart of non-TX and TX types.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery.- Preparation for delivery shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes.- The notes included in Specification MIL-S-19500, with the following exceptions or additions, are applicable to this specification.

6.2 Application guidance.- To insure proper equipment-circuit application, particular attention should be given to the differential voltage ratings and derating requirements pertinent to the individual device types covered herein.

6.3 Ordering data.-

a. Terminal-lead length: See 3.3.2 herein.

b. Lot-inspection data: See 4.2.7 herein.

6.4 Procurement of "TX" types.- See 4.2.2 herein.

6.5 Quality assurance for "TX" types.- It is expected that where the permanent lot record, maintained as required in 4.4.1 herein, begins to show a condition of failures in excess of the LTPD rate specified herein, or a condition of entire-lot rejection (see 4.4.6.4 herein) is recorded, the Government procuring activity will determine any additional monitoring controls and evaluation criteria needed to attain restoration of the prescribed (herein) quality assurance for acceptance of product.

6.6 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Commanding General U. S. Army Electronics Command, Fort Monmouth, New Jersey 07703, Attention: AMSEL-PP-EM-2.

Custodian:
Army-EL

Preparing activity:
Army-EL

Project No. 5961-A140

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-19500/389(EL) SEMICONDUCTOR DEVICE, DIODE, SILICON, HIGH-VOLTAGE RECTIFIER TYPES 1N5181, 1N5182, 1N5183, 1N5184 and TX 1N5181, TX 1N5182, TX 1N5183, TX 1N5184

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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U.S.Army Electronics Command
ATTN: AMSEL-TD-SS
Fort Monmouth, New Jersey 07703

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